WAIRAKA MODERN SS-JINJA

S.1 END OF TERM ONE 2026

Uganda Certificate of Education

PHYSICS 1 (Theory)

Time: 2 hours 15 minutes

INSTRUCTIONS

This paper has **three** items

All items are compulsory

Show your workings clearly where necessary using illustration with pen only

Use of non-programmable calculator is allowed

Begin each item on a fresh page

Attempt all item in this paper

ITEM ONE

On their way home, a group of senior one student stops at a busy roadside trading Centre where they observe a mechanic repairing a motorcycle, a bicycle repairer fixing a tyre, a technician working on a radio, and a builder measuring a glass window for fitting. The students notice that each worker is using tools and skills related to science in their daily work. As they watch, the builder carefully measures the glass while bicycle repairer checks the inside of the tire, raising curiosity among the students about how such measurements are done. Suddenly one student almost gets injured after carelessly stepping on a tool, prompting a warning from the mechanic about safety working environments. This experience makes the students begin to connect what they have seen with what they expect to learn in physics

Task:

As a student of physics,

a) Identify the different careers observed in the scenario and explain how they are related to physics

b) State the most suitable instruments you would use to measure:

i) The breadth of a glass window

ii) The internal diameter of the bicycle tyre

c) i) suggest the workshop rules that should be followed in such environments

ii)explain why following these rules is important

ITEM TWO

A miner traveling from **Mbarara** to **Kampala After finishing his business** discovers a small, precious stone along the way. Upon arriving in Kampala, he learns the metal is valued at UGX 4,100 per gram. However, he is unsure of its density and total cost and he doesn't have a weighing scale so he is advice to use water displacement method. After finishing his business, he travels back towards **Masaka City** to meet an expert who can help him verify his findings.

Hints: The actual mass of the stone is **0.0251kg** and the volume of water displaced is **250 cm^3**

The distance from Mbarara to Kampala is **500km** and the miner bus travel at a constant speed of **125km/h.**

Task:

- **a)** Describe a simple method how you will help him to determine the density of a stone he discovered (include illustration and finding to standard form in g/cm^3)
- **b)** What will be the total cost of a stone?
- **c)** Estimate the time spent from Mbarara to Kampala in
  - i) Seconds
  - ii) Minutes
- **d)** Estimate the distance in centimeters of Kampala to Masaka City

ITEM THREE

A student is provided with a pen, a bottle of juice, and a sealed tube containing iodine vapor. The student also has a fire flame and a cold surface. The student was surprised to see that when the hot iodine vapor in the tube touches the verry cold surface, it turns immediately into solid crystals, skipping the liquid phase entirely.

They also observe smoke from the flame and are told it contains tiny particles whereby he was advice to use microscope and a source of light to see tiny particles.

Task:

- **a)** Group the materials provided according to their state of matter with evidence
- **b)** Name the process responsible for the change of the hot iodine vapor in the tube to solid directly
- **c)** Describe how the student will set an experiment to demonstrate that smoke contains tiny particles (include illustration)
- **d)** Explain what would happen to the motion of the particles seen under the microscope if the temperature of the smoke cell is significantly raised using fire flame

THE END